

Shubh Gupta

+91 8120839666

✉ shubhgupta.work12@gmail.com

🌐 [LinkedIn](#)

🐙 [GitHub](#)

EDUCATION

Vellore Institute of Technology

Integrated M.Tech in Artificial Intelligence

CGPA: 8.74

October 2022 – May 2027

INTERNSHIP EXPERIENCE

Technical Solution Engineer Intern

Hyderabad, Telangana (OnSite)

AlgoUniversity (YC S21)

June 2026 – Present

- Built and evolved Archie's core AI interview engine with multi-turn orchestration, RAG-based intelligence, and canvas-aware architecture analysis for contextual follow-ups and structured feedback.
- Architected production AI/backend infrastructure using FastAPI, PostgreSQL, and a provider-agnostic LLM layer across Gemini, OpenRouter, and NVIDIA NIM for persistent interview workflows.
- Expanded Archie into a production-ready SaaS with daily credit infrastructure, Free/Pro plan management, and payment/subscription workflows for AI usage and monetization.

AI Intern – Information Technology Department

Satna, Madhya Pradesh (OnSite)

MP Birla Group (Birla Corporation Limited)

October 2025 – Dec 2025

- Built a Python + OpenCV + Tesseract system to automate form digitization, syncing extracted data to Google Sheets and reducing manual data entry effort by **85%**.
- Developed a Streamlit-based review tool for correcting OCR errors, ensuring data accuracy before final submission.
- Achieved **92% extraction accuracy** on filled forms through image preprocessing pipelines and iterative model refinement.

PROJECTS

Verba | Django, Pinecone, Gemini 2.0, Groq LLaMA 3, Supabase | [GitHub](#) | [Live Demo](#)

Jan 2026

- Built a production-ready RAG pipeline supporting PDF, DOCX, XLSX, CSV, and scanned images — using Gemini 2.0 Flash Vision OCR, Google Generative AI embeddings, and Pinecone serverless vector storage for semantic retrieval.
- Engineered a parallel OCR pipeline using **ThreadPoolExecutor** achieving **5x faster** scanned PDF processing, with async background ingestion to eliminate server timeouts and seamless UI auto-polling.
- Integrated Supabase for PostgreSQL storage, S3 file storage, and Google OAuth authentication; deployed on Render with Django 5 + Gunicorn and smart rate limiting (50 messages per 3-hour rolling window).

Deep Image Steganography | Python, TensorFlow, PyTorch, OpenCV | [GitHub](#) | [Live Demo](#)

April 2025

- Developed a generative AI-based image steganography system using GAN, VAE, and DDPM models to securely hide images within cover images with minimal visual distortion.
- Built encoder-decoder and diffusion pipelines for high-fidelity steganographic embedding and extraction of hidden data.
- Improved model robustness by **18%** using adversarial loss tuning and diffusion-based denoising for secure covert communication.

TECHNICAL SKILLS

Programming: Python, C++, SQL

Coursework: Data Structures & Algorithms, DBMS, Computer Networks, Operating System

AI & Machine Learning: Deep Learning, GANs, NLP, Computer Vision, Generative AI, RAG

Frameworks & Libraries: PyTorch, TensorFlow, Keras, OpenCV, FastAPI, Django

Tools & Platforms: Git, GitHub, Docker, Pinecone, Supabase, Tesseract OCR, Google Colab, VS Code

ACHIEVEMENTS

- **NASSCOM AI Hackathon:** Semi-finalist
- **TATA Imagination Challenge 2025:** Finalist